

Information Security

Bonus Assignment 2

You are restricted to the following tools. Any additional tool requires prior approval from the Course Instructor and TA via email.

Tools:

- x64dbg (for manual patching)
- C2 framework of your choice
- Shellcode generator of your choice
- Target software of your choice (must not be your own program)
- Microsoft Windows (10/11)

Automated patchers are not allowed.

Reverse Engineering, Static Patching & C2

Question 1 (4 Points)

Select a small Windows application (not self-written). Perform static patching to embed a simple proof-of-concept shellcode directly inside the executable.

Requirements:

- Shellcode must reside inside the executable (no external file calls).
- Do not modify the source code; use reverse engineering methodology only.
- You are not allowed to modify the Original Entry Point (OEP).
- You must patch a function or hijack an imported function.
- You may use a code cave and safely redirect control flow (recommended).
- The patched executable must remain functional.
- Shellcode must perform a visible action (e.g., launch calculator, notepad, etc.).
- Shellcode for Question 1 must be different from Question 2.

A patched executable must be produced.

Question 2 (6 Points)

Select a C2 framework of your choice (e.g., Sliver, Covenant, Havoc, Mythic, or others). Generate reverse connection shellcode (e.g., reverse TCP) for your C2 server. Patch the same or another selected software with the generated shellcode.

Requirements:

- Static patching only (manual using x64dbg).
- No automated patching tools.
- No modification of the OEP.
- Patch must hijack an internal function or imported function.
- The executable must remain functional.

- Provide proof of maintained C2 connection upon execution.

Submission Requirements

Submit:

1. A structured PDF report (not compressed) containing:
 - Step-by-step explanation
 - Corresponding screenshots for each step
2. A compressed file (.zip/.rar) containing:
 - Two working patched executables (one for Q1 and one for Q2)

Must followed Pattern in PDF report:

Explanation
Picture

Note: You are not required to write a full program. Shellcode generation is required. All patching must be performed manually using a reverse engineering tool.